Introduction to Quantum Physics and Structure of Matter

Lecture 6

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

Lecture	Date	Topics	Material
1	03-09-2025	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	10-09-2025	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	17-09-2025	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	24-09-2025	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

Lecture	Date	Topics	Material
5	01-10-2025	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	08-10-2025	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	**15-10-2025**	**•Nuclear constituents** **•Nuclear sizes and shapes** **•Alpha decay** **•Beta decay** **•Gamma decay and nuclear excited states**	**Book: Ch. 12.1-12.3** **12.6-12.9**
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

Recap Previous Lecture

- Quantum numbers & Pauli principle

No two electrons in a single atom can have the same set of quantum numbers: (n, l, m_l, m_s)

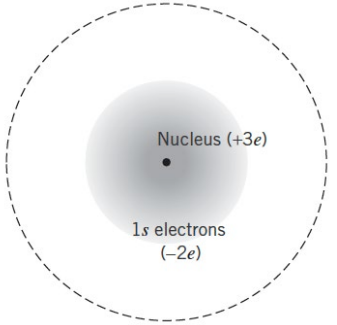
- Atomic shells and subshells

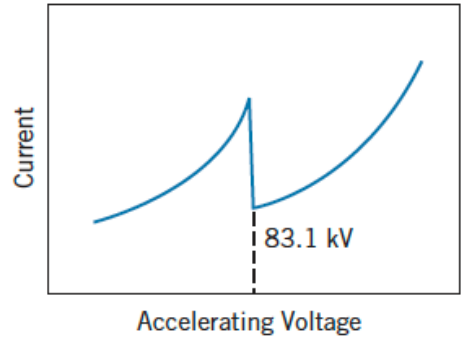
n	1	2	3
Shell	K	L	M

TABLE 8.1 Filling of Atomic Subshells

n	l	Subshell	Capacity $2(2l+1)$
1	0	1s	2
2	0	2s	2
2	1	2p	6
3	0	3s	2
3	1	3p	6
4	0	4s	2
3	2	3d	10

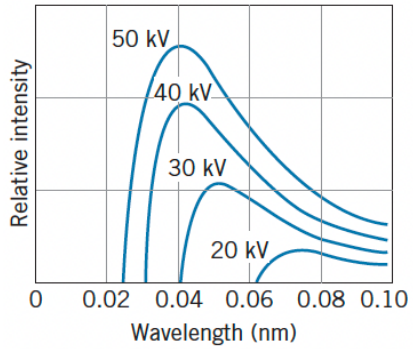
- Electron Screening & absorption edge

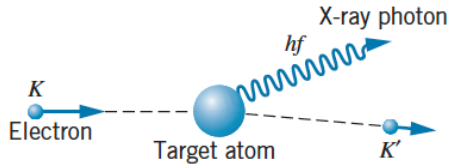




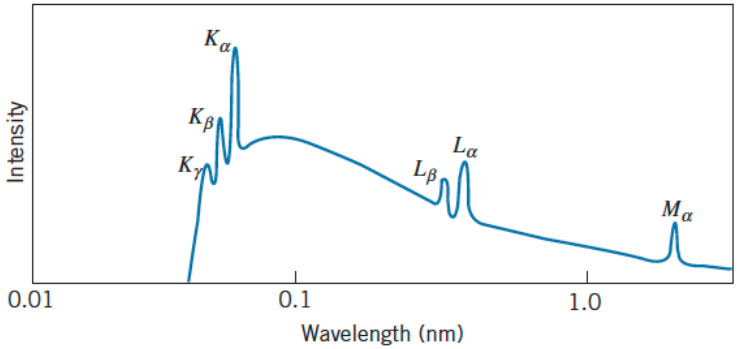
Recap Previous Lecture

- Continuous X-Ray radiation





- Characteristic X-Ray radiation



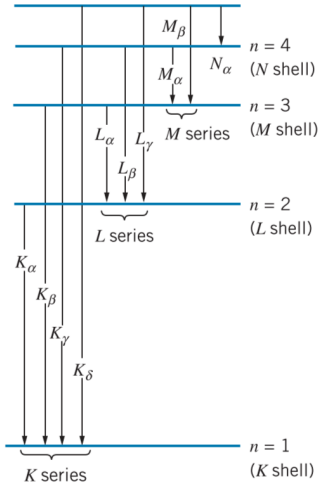


FIGURE 8.14 X-ray series.

The Atomic Nucleus



- Mass of the nucleus provides 99.9% of atomic mass
- Volume of the nucleus is 10^{-15} of atomic volume

- It has a charge of $+Ze$

- Besides Z protons, there are also N neutrons in the nucleus

$$^{A}_{Z}X_{N}$$

$A \rightarrow$ The mass number

$Z \rightarrow$ The atom number

$N \rightarrow$ The neutron number

TABLE 12.1 Properties of the Nucleons

Name	Symbol	Charge	Mass	Rest Energy	Spin
Proton	p	$+e$	1.007276 u	938.28 MeV	$^1/_2$
Neutron	n	0	1.008665 u	939.57 MeV	$^1/_2$

The Atomic Nucleus

- Nuclei do not have an easily definable radius or surface

- The density of a nucleus is independent on the mass number A

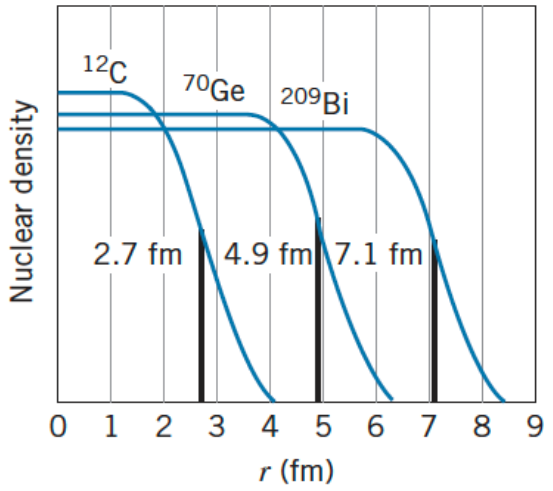


FIGURE 12.1 The radial dependence of the nuclear charge density.

Assuming the nucleus is a 'sphere' with radius R:

$$\frac{\text{number of neutrons and protons}}{\text{volume of nucleus}} = \frac{A}{\frac{4}{3}\pi R^3} \cong \text{constant}$$

$$\Rightarrow R = R_0 A^{1/3} \qquad \text{with} \qquad R_0 \approx 1.2 \cdot 10^{-15}\text{m} = 1.2\ \text{fm}$$

TU/e

The Atomic Nucleus – Exercise

Compute the density of a typical nucleus, and find the resultant mass if we could produce a nucleus with a radius of 1 cm

$$R = R_0 A^{1/3} \quad \text{with} \quad R_0 \approx 1.2 \cdot 10^{-15}\text{m} = 1.2\ \text{fm}$$

The Atomic Nucleus – Exercise

Compute the density of a typical nucleus, and find the resultant mass if we could produce a nucleus with a radius of 1 cm

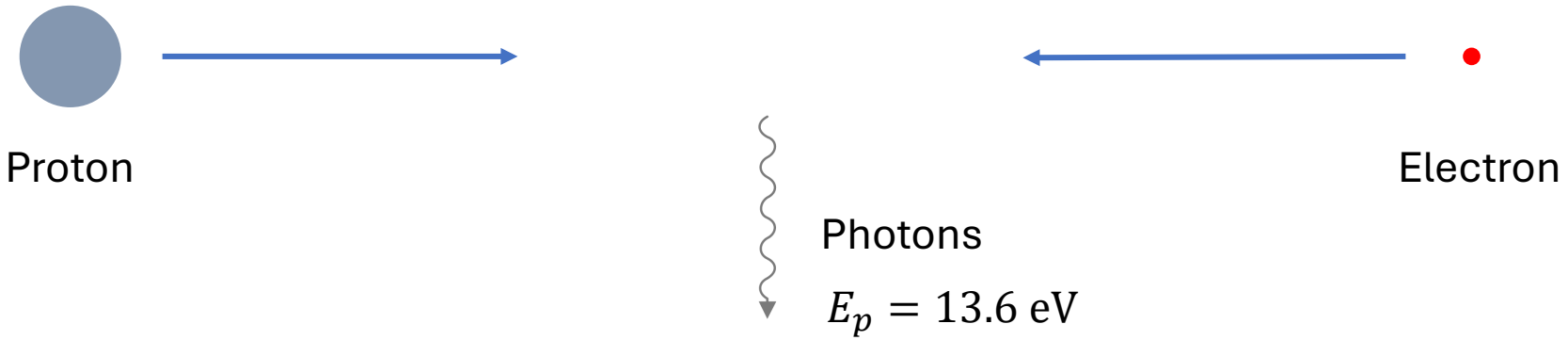
$$R = R_0 A^{1/3} \quad \text{with} \quad R_0 \approx 1.2 \cdot 10^{-15}\text{m} = 1.2\ \text{fm}$$

$$\rho = \frac{m}{V} = \frac{Am_P}{\frac{4}{3}\pi R^3} = \frac{Am_P}{\frac{4}{3}\pi R_0^3 A} = \frac{m_P}{\frac{4}{3}\pi R_0^3} = \frac{(1.67 \cdot 10^{-27})}{\frac{4}{3}\pi(1.2 \cdot 10^{-15})} = 2 \cdot 10^{17}\ \text{kg/m}^3$$

The mass of a hypothetical 1 cm nucleus would be:

$$m = \rho V = (2 \cdot 10^{17})\left(\frac{4}{3}\pi \cdot 0.01^3\right) = \mathbf{8 \cdot 10^{11}\ kg}$$

Making Hydrogen



Conservation of energy:

$$m_e c^2 + m_p c^2 = m(\mathrm{H})c^2 + 13.6\ \mathrm{eV}$$

$$m_e c^2 + m_p c^2 - m(\mathrm{H})c^2 = \boxed{13.6\ \mathrm{eV}} \rightarrow$$

Binding energy of the atom

TU/e

Binding Energy

- We can also calculate the binding energy for a nucleus:

- For a nucleus of atom ${}^{A}_{Z}X_{N}$ with mass m_X we have:

$$B = Nm_n c^2 + Zm_p c^2 - m_X c^2$$

- Typically atomic, not nuclear mass is measured. Relationship between atomic & nuclear mass:

$$m\left({}^{A}_{Z}X_{N}\right)c^2 = m_X c^2 + Zm_e c^2 - \cancel{B_e}$$

with B_e the binding energy of the electrons, may be ignored (order of magnitude $10^0 - 10^2$ eV, $\ll m_X c^2$ and $Zm_e c^2$)

Binding Energy

Using:

nuclear mass

$$B = Nm_n c^2 + Zm_p c^2 - m_X c^2$$

$$m\left({}^{A}_{Z}X_N\right)c^2 = m_X c^2 + Zm_e c^2 \quad \text{(neglecting } B_e\text{)}$$

Atomic mass

$$m\left({}^{1}_{1}\mathrm{H}\right)c^2 = m_p c^2 + m_e c^2 \quad \text{(neglecting } B_e\text{)}$$

We find:

$$B = Nm_n c^2 + Z\left[m\left({}^{1}_{1}\mathrm{H}\right)c^2 - m_e c^2\right] - \left[m\left({}^{A}_{Z}X_N\right)c^2 - Zm_e c^2\right] \Longrightarrow$$

$$\boxed{B = \left[Nm_N + Zm\left({}^{1}_{1}\mathrm{H}_0\right) - m\left({}^{A}_{Z}X_N\right)\right]c^2}$$

Note: the electron masses, m_e, cancel in the equation

Binding Energy – Exercise

Find the total binding energy B and also the average binding energy per nucleon B/A for ${}^{56}_{26}\mathrm{Fe}_{30}$ and ${}^{238}_{92}\mathrm{U}_{146}$.

$$B = \left[Nm_N + Zm\left({}^{1}_{1}\mathrm{H}_0\right) - m\left({}^{A}_{Z}X_N\right)\right]c^2$$

	Z	A	Atomic mass (u)	*Abundance* or Half-life
Fe	26	52	51.948114	8.27 h
		53	52.945308	8.51 m
		54	53.939611	*5.85%*
		55	54.938293	2.74 y
		56	55.934937	*91.75%*
		57	56.935394	*2.12%*
		58	57.933276	*0.28%*
		59	58.934875	44.5 d
		60	59.934072	1.5 My
		61	60.936745	6.0 m

	Z	A	Atomic mass (u)	*Abundance* or Half-life
U	92	233	233.039635	0.1592 My
		234	234.040952	0.2455 My
		235	235.043930	*0.720%*
		236	236.045568	23.42 My
		237	237.048730	6.75 d
		238	238.050788	*99.274%*
		239	239.054293	23.5 m

$$M\left({}^{1}_{1}\mathrm{H}\right) = {}^{1}_{1}M = 1.007825\,u$$

Binding Energy – Exercise

Find the total binding energy B and also the average binding energy per nucleon B/A for ${}^{56}_{26}\mathrm{Fe}_{30}$ and ${}^{238}_{92}\mathrm{U}_{146}$.

$$B = \left[Nm_N + Zm\left({}^{1}_{1}\mathrm{H}_0\right) - m\left({}^{A}_{Z}X_N\right)\right]c^2$$

For ${}^{56}_{26}\mathrm{Fe}_{30}$:

$$B = [30(1.008665\ \mathrm{u}) + 26(1.007825\ \mathrm{u}) - 55.934937\ \mathrm{u}](931.5\ \mathrm{MeV/u}) = 492.3\ \mathrm{MeV}$$

$$B/A = \frac{492.3}{56} = 8.790\ \mathrm{MeV\ per\ nucleon}$$

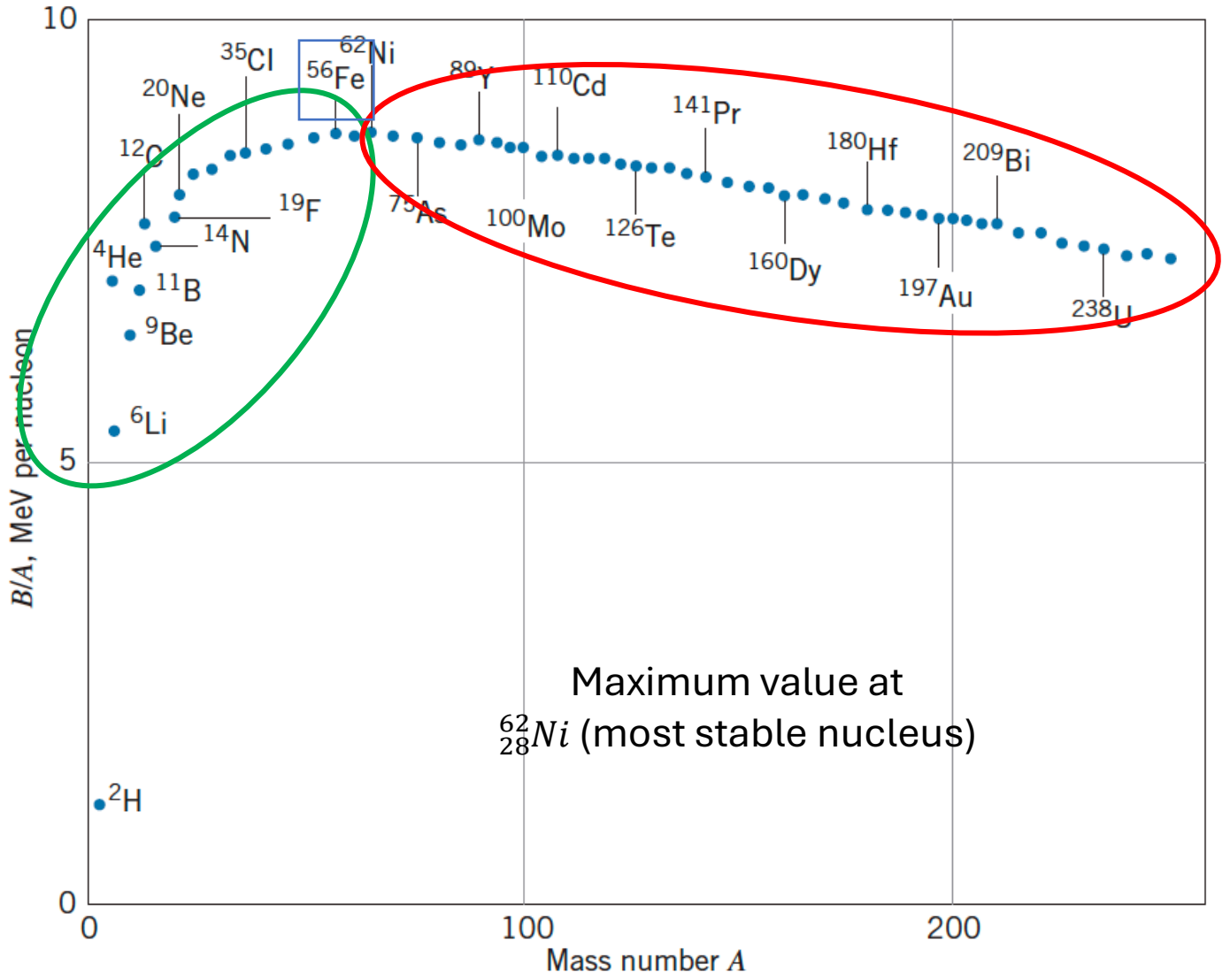
For ${}^{238}_{92}\mathrm{U}_{146}$:

$$B/A = \frac{1802}{238} = 7.570\ \mathrm{MeV\ per\ nucleon}$$

Binding Energy

$A < 56$

Nuclear Fusion: binding energy per nucleon is greater in the final nucleus than it is in the two original nuclei



$A > 56$

Nuclear Fission: splitting massive nucleus into two lighter nuclei, binding energy per nucleon is greater for the two lighter fragments than it is for the original nucleus

Ni-62 has the highest binding energy per nucleon of any known nuclide (8.7945 MeV).
It is often stated that Fe-56 is the "most stable nucleus", but this is because Fe-56 has the lowest mass per nucleon, not binding energy per nucleon, of all nuclides. (Wikipedia)

TU/e

Separation Energies

- Proton separation

$${}^{A}_{Z}X_N \rightarrow {}^{A-1}_{Z-1}X'_N + p: \qquad S_p = \left[m\left({}^{A-1}_{Z-1}X'_N\right) + m\left({}^{1}_{1}\mathrm{H}\right) - m\left({}^{A}_{Z}X_N\right)\right]c^2$$

- Neutron separation

$${}^{A}_{Z}X_N \rightarrow {}^{A-1}_{Z}X_{N-1} + n: \qquad S_n = \left[m\left({}^{A-1}_{Z}X_{N-1}\right) + m_n - m\left({}^{A}_{Z}X_N\right)\right]c^2$$

Stable and Unstable Nuclei

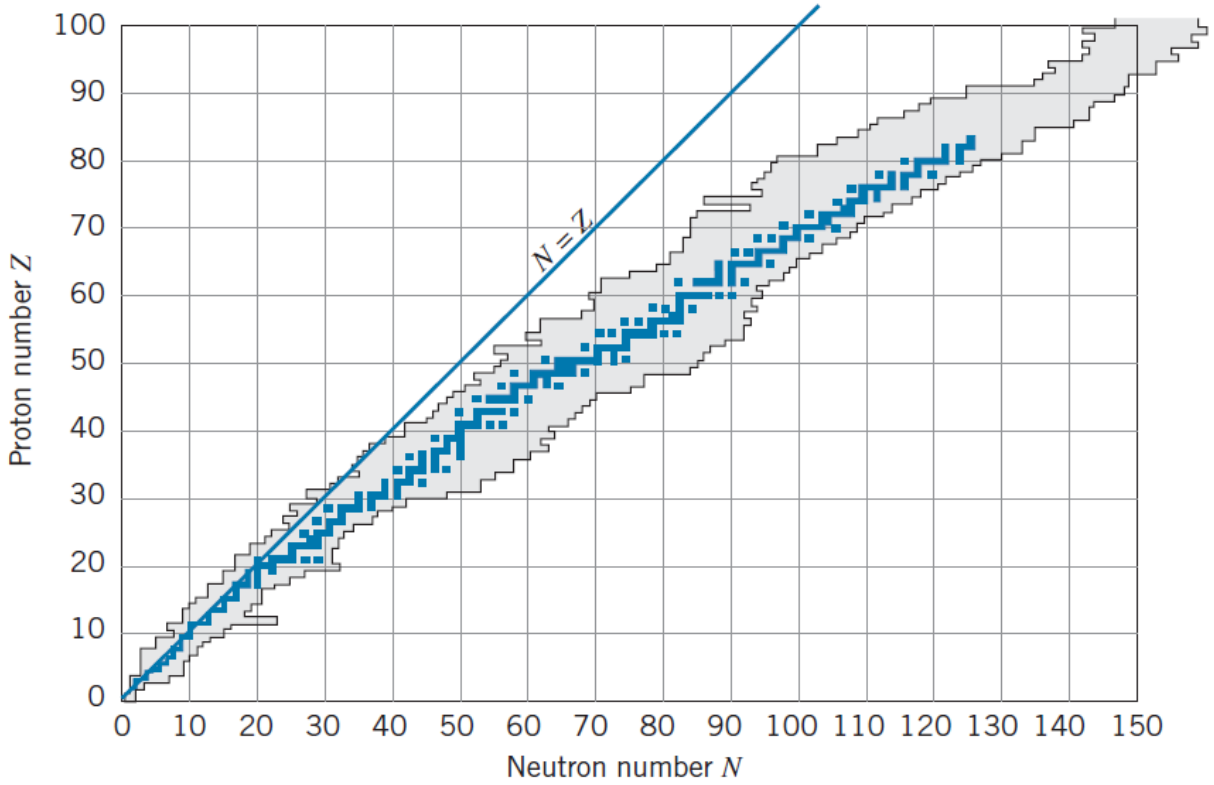
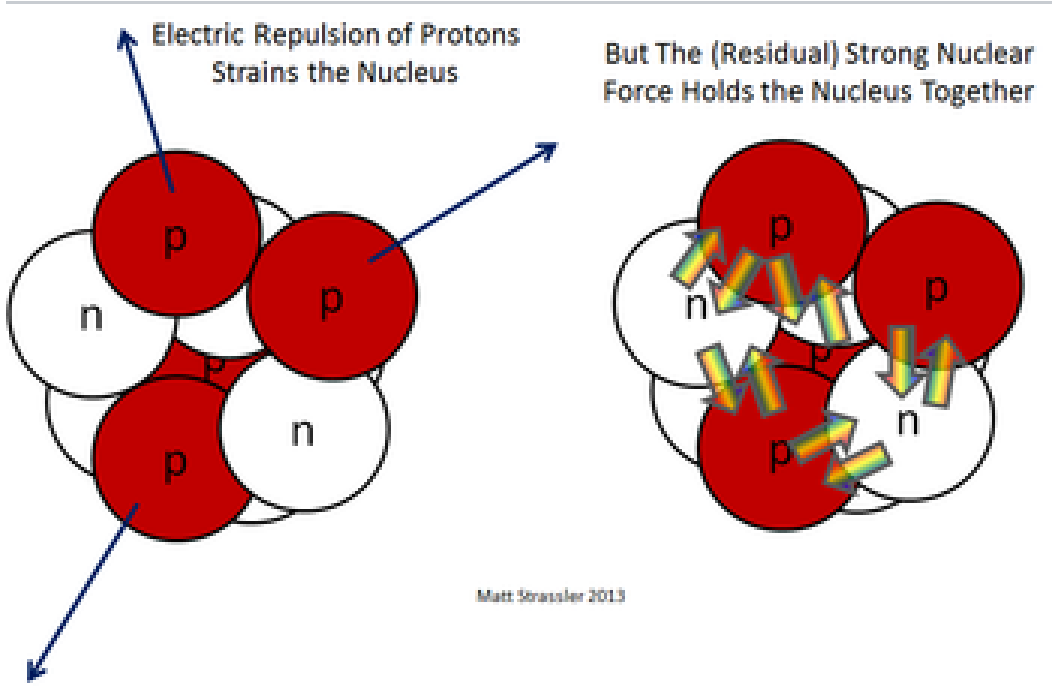


FIGURE 12.11 Stable nuclei are shown in color; known radioactive nuclei are in light shading.

- Horizontal lines: isotopes
- Large N: more neutrons are needed to hold nucleus together
- Unstable nuclei transform themselves in stable nuclei due to *radioactive decay*

TU/e

Nuclear glue



Strong force: 10-100 times stronger than EM (Coulomb) force but very short range only (~fm)

EM force : $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1 q_2}{r^2}\right) \approx \frac{1}{4\pi\epsilon_0}\left(\frac{Z(Z-1)e^2}{R^2}\right)$

TU/e

Forces on nuclear parts

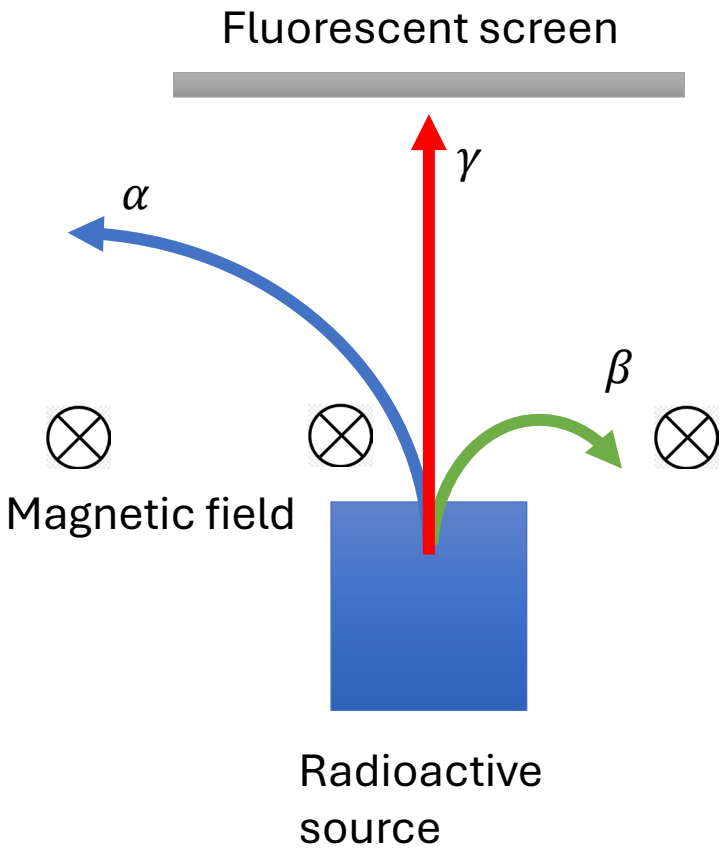
Interaction	Relative Strength	Range (m)	Particle(s) Affected	Exchange Particles	Masses of Exchange Particles (GeV/c^2)
strong	1	10^{-15} m	quarks	gluons (g)	0
electromagnetic	10^{-2}	∞	electrically charged	photon (γ)	0
weak	10^{-6}	10^{-18} m	quarks and leptons	W^+, W^-, Z^0	80.4, 80.4, 91.2
gravitational	10^{-43}	∞	all	graviton	0

Binds nuclear particles

Changes quarks:
p into n and vv)-> β radiation

The word *strong* is used since the strong interaction is the "strongest" of the four fundamental forces. At a distance of 1 femtometer (1 fm = 10^{-15} meters) or less, its strength is around 137 times that of the electromagnetic force, some 10^6 times as great as that of the weak force, and about 10^{38} times that of gravitation.

Types of radiation



Discovered by Rutherford, 1899
Various substances emit radiation:

$\boldsymbol{\alpha}$: easily stopped by sheet of paper; deflected by magnetic field
$\boldsymbol{\beta}$: stopped by thin sheet of metal; deflected by magnetic field
$\boldsymbol{\gamma}$: strong penetrating effect; Not deflected by magnetic field

Radioactive Decay

- Radioactive decay: unstable nucleus loses energy by emitting particles and/or energy

- N = number of (radioactive) nuclei
- a = activity: rate at which unstable nuclei decay

$$a = -\frac{dN}{dt}$$

- Activity is product of **decay probability λ** and number of instable nuclei:

$$a = \lambda N$$

Marie Curie
1867-1934



$$1\ \text{curie (Ci)} = 3.7 \cdot 10^{10}\ \frac{\text{decays}}{\text{s}}$$

$$1\ \text{Ci} = 3.7 \cdot 10^{10}\ \text{(Becquerel)Bq}$$

Radioactive Decay

$$a = \lambda N = -\frac{dN}{dt} \quad \Rightarrow \quad \frac{dN}{N} = -\lambda dt$$

Integration gives us:

$$N(t) = N_0 e^{-\lambda t}$$

We cannot count N, but we measure activity

$$\lambda N(t) = a(t) = a_0 e^{-\lambda t}$$

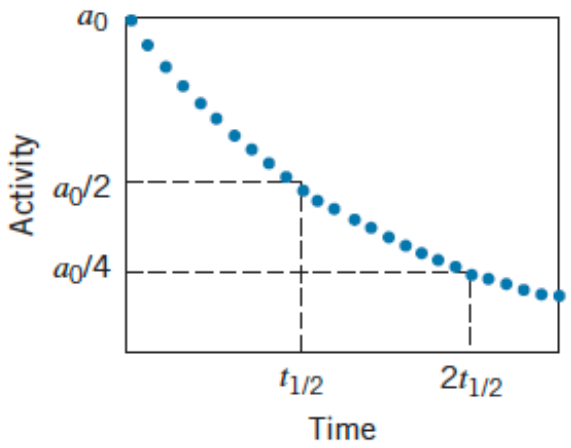
Radioactive Decay

- **Half-time $t_{1/2}$**: "The time it takes for half of the atoms of a radioactive material to decay/for the activity level to drop to half the original level"

$$N(t = t_{1/2}) = N_0 e^{-\lambda t_{1/2}} = \frac{1}{2} N_0 \quad \text{or} \quad a(t = t_{1/2}) = a_0 e^{-\lambda t_{1/2}} = \frac{1}{2} a_0$$

$$e^{-\lambda t_{1/2}} = \frac{1}{2}$$

$$t_{1/2} = -\frac{\ln 1/2}{\lambda} = \frac{\ln 2}{\lambda}$$



- **Mean lifetime τ**: $\tau = \frac{1}{\lambda}$

TU/e

Radioactive Decay – Exercise

The half-life of $^{198}\mathrm{Au}$ is 2.70 days.

(a) What is the decay constant of $^{198}\mathrm{Au}$?

(b) What is the probability that any $^{198}\mathrm{Au}$ nucleus will decay in one second?

(c) Suppose we had a $1.00\mu\mathrm{g}$ sample of $^{198}\mathrm{Au}$. What is its activity?

(d) How many decays per second occur when the sample is one week old?

Radioactive Decay – Exercise

a) $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{2.70\ \mathrm{d}} = \frac{0.693}{2.70 \times 24 \times 3600} = 2.97 \cdot 10^{-6} \mathrm{s}^{-1}$

b) This is the decay constant: one nucleus per $2.97 \cdot 10^{-6} \mathrm{s}$

c) $N = \frac{mN_A}{M} = \frac{(1.00 \cdot 10^{-6})(6.02 \cdot 10^{23})}{198} = 3.04 \cdot 10^{15}$ atoms

$a = \lambda N = (2.97 \cdot 10^{-6})(3.04 \cdot 10^{15}) = 9.03 \cdot 10^{9}\ \mathrm{Bq} = 0.244\ \mathrm{Ci}$

d) $a = a_0 e^{-\lambda t} = (9.03 \cdot 10^{9})\, e^{-(2.97 \cdot 10^{-6})(7 \times 24 \times 3600)} = 1.50 \cdot 10^{9}\ \mathrm{Bq}$

Conservation Laws

When particles are created and/or emitted, conservation laws apply:

- Energy
- Linear momentum
- Angular momentum (orbital + spin)
- Electric charge
- Nucleon number

The Q-value

- Excess rest energy after decay is determined by the Q-value:

$$X \rightarrow X' + x$$

$$Q = \left(m_X - \left(m_{X'} + m_x\right)\right) c^2$$

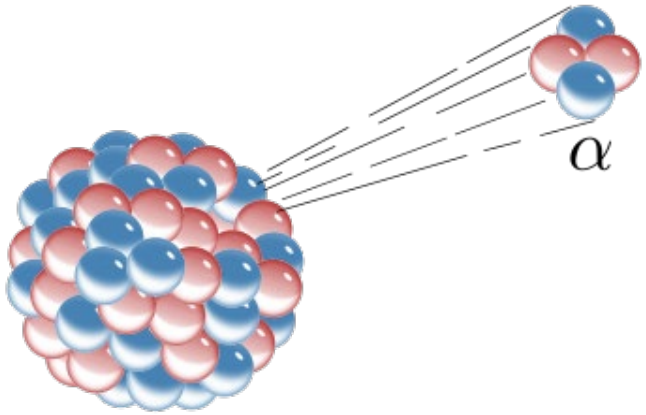
where the m's are nuclear masses. As before, electron masses cancel out, so we can use atomic masses

- Q-value is the difference between initial and final nuclear rest energies
  - $Q > 0$: exotherm, decay is possible
  - $Q < 0$: endotherm, decay not possible

Alpha Decay

- Parent X, daughter X':

$$ {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X' + \alpha $$



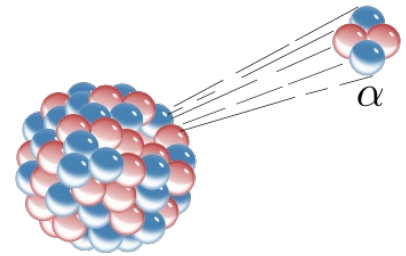
- α radiation consists of particles: $2p + 2n$ or Helium nuclei (${}^{4}_{2}\mathrm{He}_{2}$)

- Heavy particle, can be easily stopped (sheet of paper)
- Can ionize fairly easily due to high mass- dangerous when ingested

- Decay process releases energy ($Q > 0$)

Alpha Decay

$$ {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X' + \alpha $$



e.g. ${}^{226}_{88}Ra \rightarrow {}^{224}_{86}Rn + \alpha$

- Q-value:

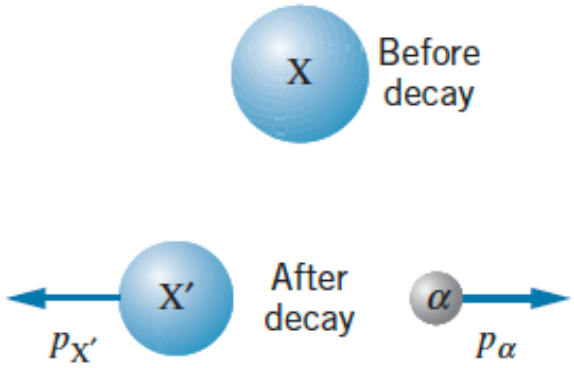
$$ Q = \left[m(X) - m(X') - m\left({}^{4}_{2}\mathrm{He}\right)\right]c^2 $$

- The released energy appear as kinetic energy of the particles:

$$ Q = K_{X'} + K_{\alpha} $$

- Conservation of linear momentum:

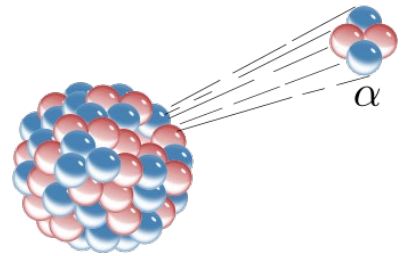
$$ p_{X'} = p_{\alpha} $$



TU/e

Alpha Decay

$$ {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X' + \alpha $$



Energies only few MeV, can use non-relativistic K, p:

$$ K = \frac{p^2}{2m} \quad p_{X'} = p_\alpha = p $$

$$ K_\alpha = \frac{p^2}{2m_\alpha} \qquad K_{X'} = \frac{p^2}{2m_{X'}} $$

$$ Q = K_\alpha + K_{X'} = \frac{p^2}{2}\left[\frac{1}{m_\alpha} + \frac{1}{m_{X'}}\right] $$

$$ Q \cdot \left[\frac{\frac{1}{m_\alpha}}{\frac{1}{m_\alpha} + \frac{1}{m_{X'}}}\right] = \frac{p^2}{2m_\alpha} = K_\alpha $$

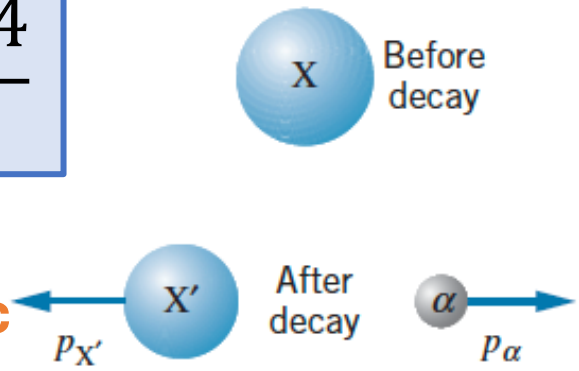
$$ K_\alpha = Q \cdot \left[\frac{\frac{1}{m_\alpha}}{\frac{m_\alpha + m_{X'}}{m_\alpha m_{X'}}}\right] = Q \cdot \frac{m_{X'}}{m_\alpha + m_{X'}} $$

Using $m_\alpha \approx 4u$ and $m_{X'} \approx (A-4)u$:

$$ K_\alpha \cong Q \cdot \frac{(A-4)u}{4u + (A-4)u} = Q \cdot \frac{A-4}{A} $$

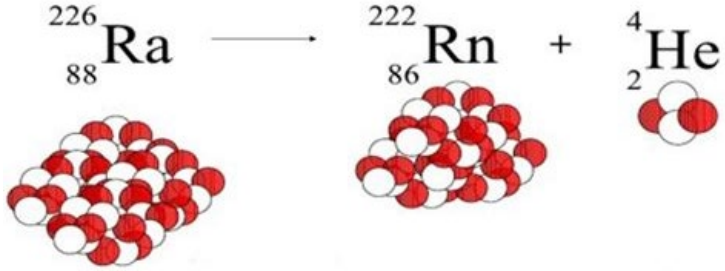
$$ K_\alpha \cong Q \cdot \frac{A-4}{A} $$

α particle is **mono-energetic**



TU/e

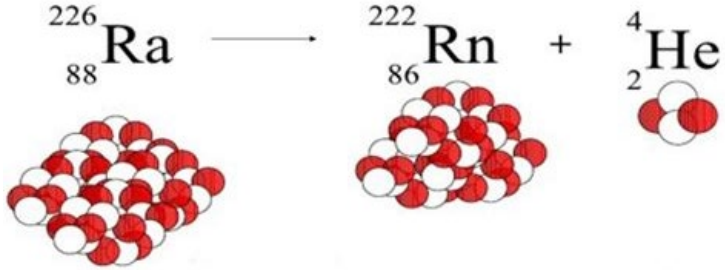
Alpha Decay – Exercise



Find the kinetic energy of the alpha particle emitted in the alpha decay process ${}^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$.

Use Krane appendix D for atomic masses!

Alpha Decay – Exercise



Find the kinetic energy of the alpha particle emitted in the alpha decay process $^{226}\mathrm{Ra} \rightarrow {}^{222}\mathrm{Rn} + {}^{4}\mathrm{He}$.

$$Q = \left[m\left(^{226}\mathrm{Ra}\right) - m\left(^{222}\mathrm{Rn}\right) - m\left(^{4}\mathrm{He}\right)\right]c^2 =$$
$$(226.026097\ \mathrm{u} - 222.017576\ \mathrm{u} - 4.002603\ \mathrm{u})(931.5\ \mathrm{MeV/u})$$
$$= 4.872\ \mathrm{MeV}$$

$$K_\alpha = \frac{A-4}{A}Q = \left(\frac{222}{226}\right)(4.872) = 4.785\ \mathrm{MeV}$$

TU/e

Alpha Decay

- Alpha decay is an example of quantum-mechanical *tunneling*
- The total potential is the sum of attracting nucleus (strong force) and repulsive Coulomb force
- Energy: $\sim 1 - 4$ MeV, happens mostly for heavy nuclei ($A > 150$)

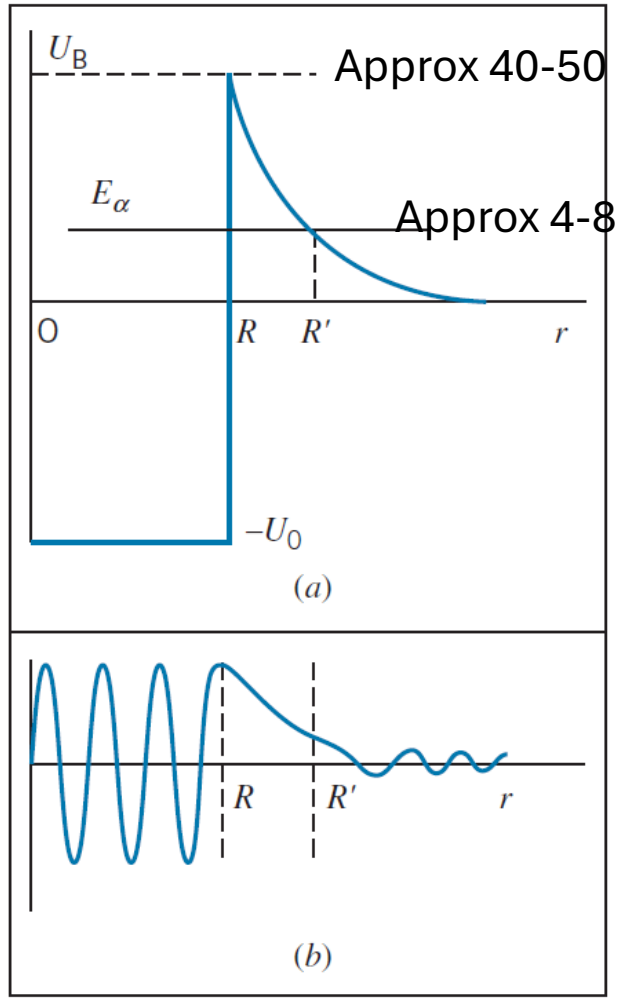


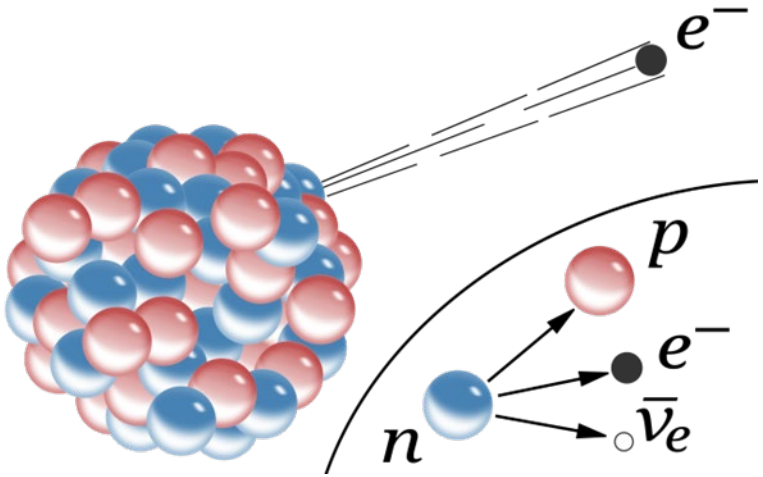
FIGURE 12.15 (*a*) The potential energy barrier for an alpha particle. (*b*) A representation of the wave function of the alpha particle.

TU/e

Beta Decay

- Beta radiation (β^- radiation): consists of (fast-moving) electrons
- More penetrating power than α radiation (stopped by thin sheet of aluminium)
- Variable energy
- Shield with low-Z materials (e.g. perspex) to avoid Bremsstrahlung

Beta Decay – (anti)neutrinos



Inside nucleus: **neutron decay:** $n \rightarrow p + \beta^-$

But:

angular momentum not conserved; energy not mono-energetic

Pauli(1930) postulates existance of additional particle:

neutrino ($\bar{\nu}$)

$$n \rightarrow p + \beta^- + \bar{\nu}$$

(Exprimentally confirmed 1956)

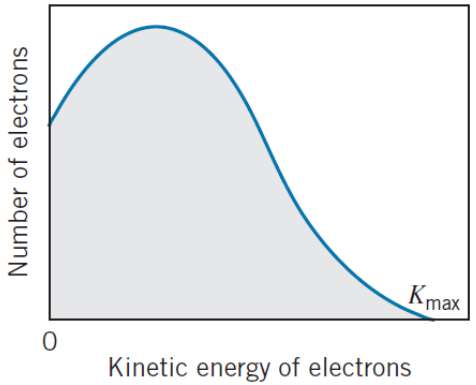


FIGURE 12.17 Spectrum of electrons emitted in beta decay.

Satisfies conservation of:

- **Energy:** β^- must share its energy with another particle (that has no/very little mass)
- **Angular momentum:** particle must have spin $\pm\frac{1}{2}$
- **Electric charge:** particle must be electrically neutral (hence 'neutrino')

TU/e

Beta Decay

β^- via neutron decay:

$\bar{\nu}$ mass negligible

$$ {}^{A}_{Z}X_N \rightarrow {}^{A}_{Z+1}X'_{N-1} + e^- + \bar{\nu} $$

$$ Q = \left[m\left({}^{A}X\right) - m\left({}^{A}X'\right) \right] c^2 $$

Electron masses cancel out, can use atomic masses

Q distributed almost exclusively accross $e^-, \bar{\nu}$

Beta Decay – Exercise

$^{23}\mathrm{Ne}$ decays to $^{23}\mathrm{Na}$ by negative beta (β^-) emission. What is the maximum kinetic energy of the emitted electrons?

Beta Decay – Exercise

$^{23}\mathrm{Ne}$ decays to $^{23}\mathrm{Na}$ by negative beta emission. What is the maximum kinetic energy of the emitted electrons?

$$^{23}\mathrm{Ne} \rightarrow {}^{23}\mathrm{Na} + e^{-} + \bar{\nu}$$

$$Q = \left[m\left(^{23}\mathrm{Ne}\right) - m\left(^{23}\mathrm{Na}\right)\right]c^{2}$$
$$= (22.994467\ \mathrm{u} - 22.989769\ \mathrm{u})(931.5\mathrm{MeV/u}) = 4.376\ \mathrm{MeV}$$

Neglecting the small correction for the kinetic energy of the recoiling nucleus, the maximum kinetic energy of the electrons is equal to this value (which occurs when the neutrino has a negligible energy).

Beta Decay –Clinical use

- Betas have limited (mm) penetrating power, so perfect for localized treatment

- Radioactive source in or nearby target tissue, high dose of radiation deposited only nearby the source

Slides and images from Catharina Hospital

Beta min decay

- Example complex beta emitter (I-131 decay to Xe-131):

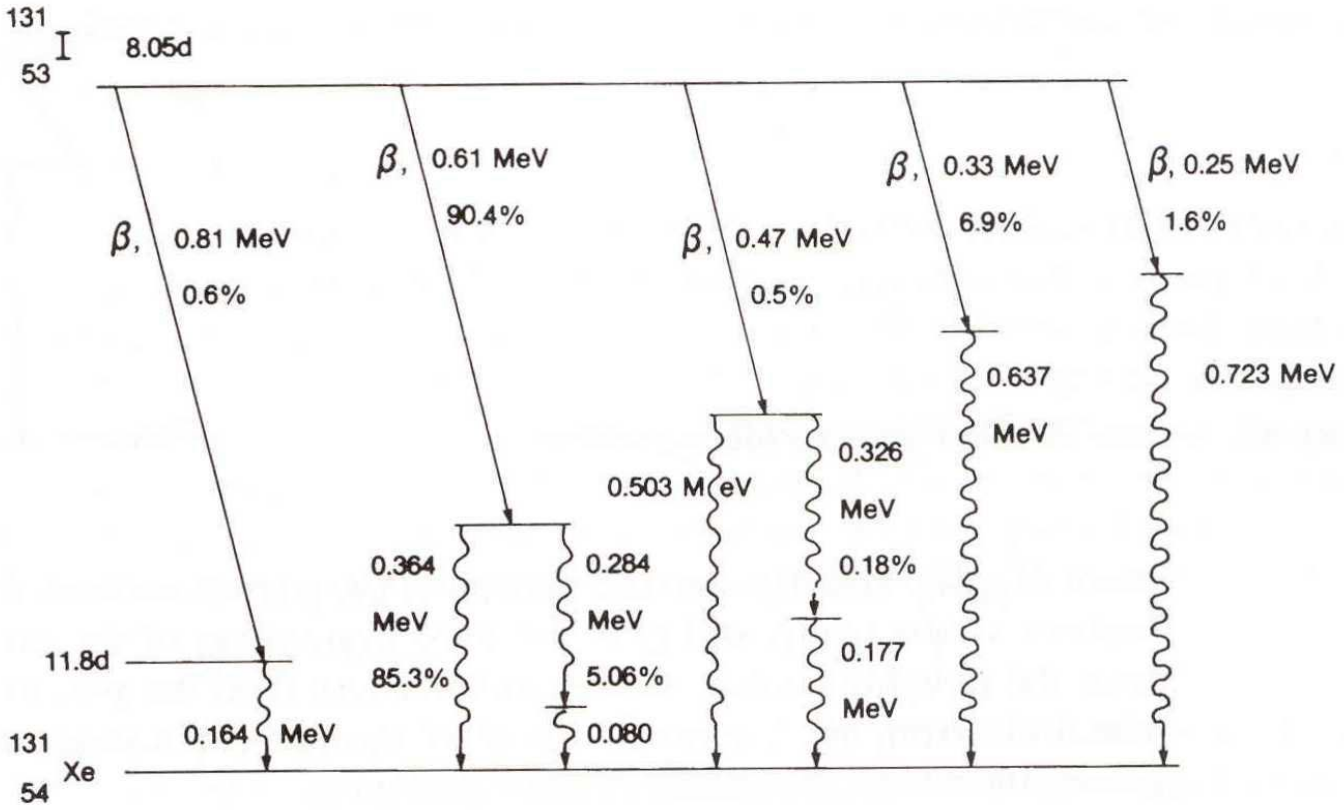
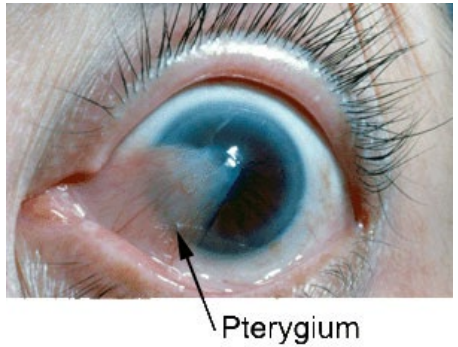
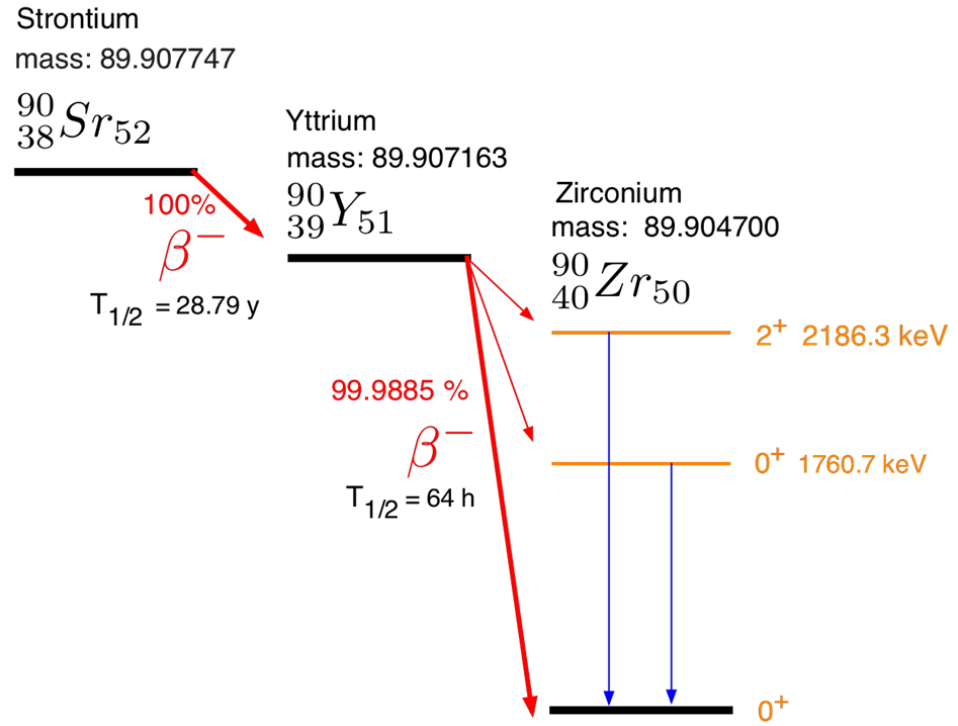


FIGURE 4.7. Iodine-131 transformation (decay) scheme.

Beta Decay – Examples from the Clinic



- Pterygium: Winglike membrane growth over the eye (“surfers’s eye”) irradiated with β^- source ^{90}Sr



Slides and images from Catharina Hospital

Beta Decay: β^+ betas

- Proton decay: $p \rightarrow n + e^+ + \nu$

- Releases anti-electron or *positron* (e^+ or β^+)
- Q value for free protons < 0 , free proton decay (H atoms) not observed
- Proton decay is possible in some nuclei:

$$ {}^{A}_{Z}X_N \rightarrow {}^{A}_{Z-1}X'_{N+1} + e^+ + \nu $$

$$ Q = \left[m\left({}^{A}X\right) - m\left({}^{A}X'\right) - 2m_e \right] c^2 $$

β^+ decay – clinical use

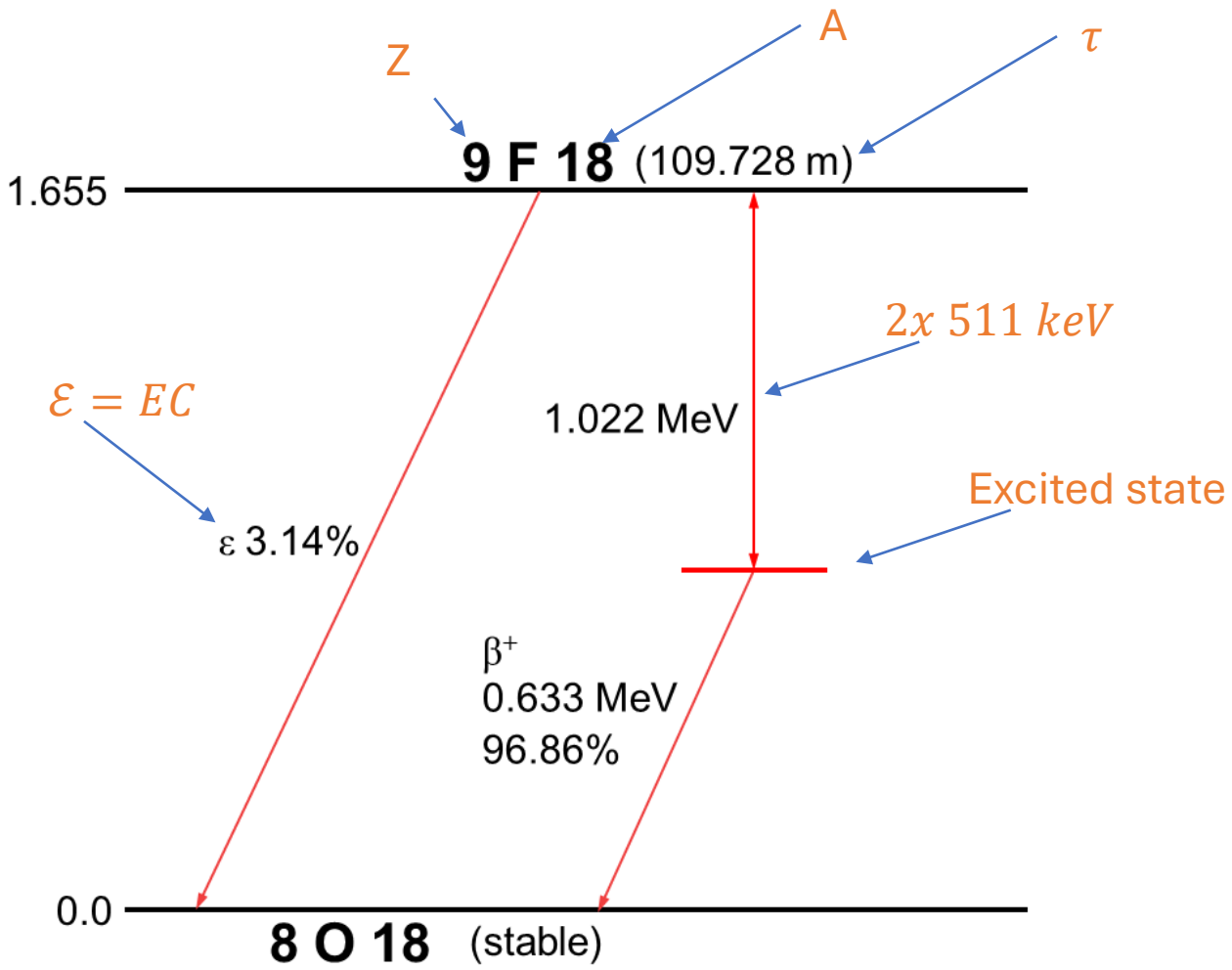
- Positron (or β^+, or e^+) is *anti-matter:*

- After slowing down of the positron it finds an electron and the two **annihilate**, resulting in two gamma's, each with an energy of $m_e = 511\ keV$ and travelling in opposite directions (conservation of energy and momentum)

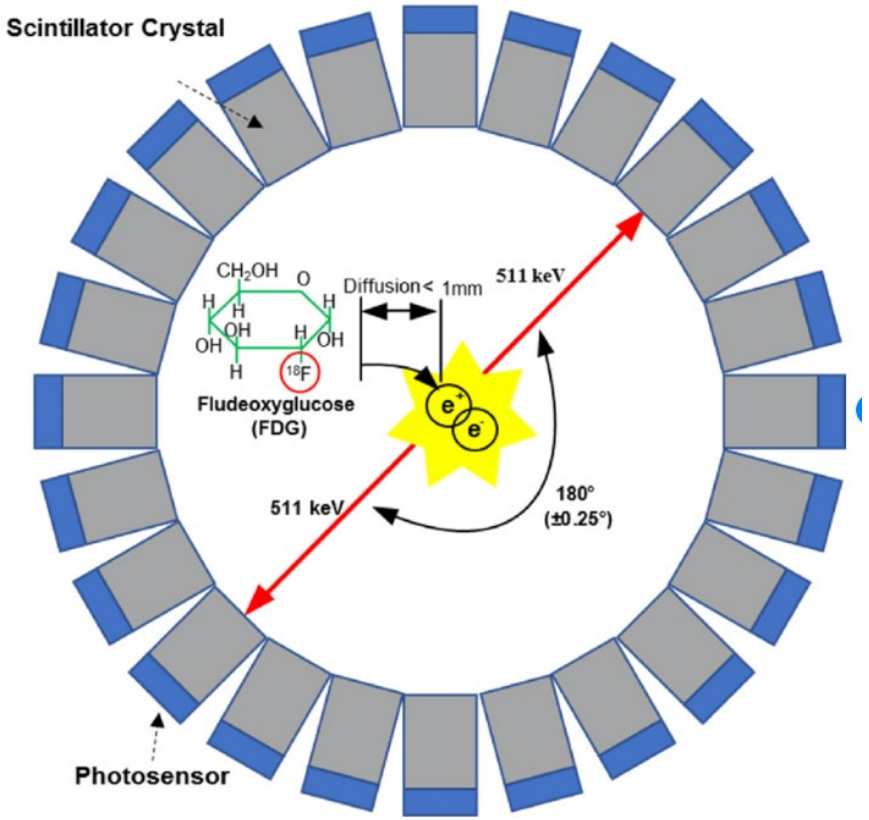
- Basis for PET (positron emission tomography)

Example from the clinic: ^{18}F in FDG PET

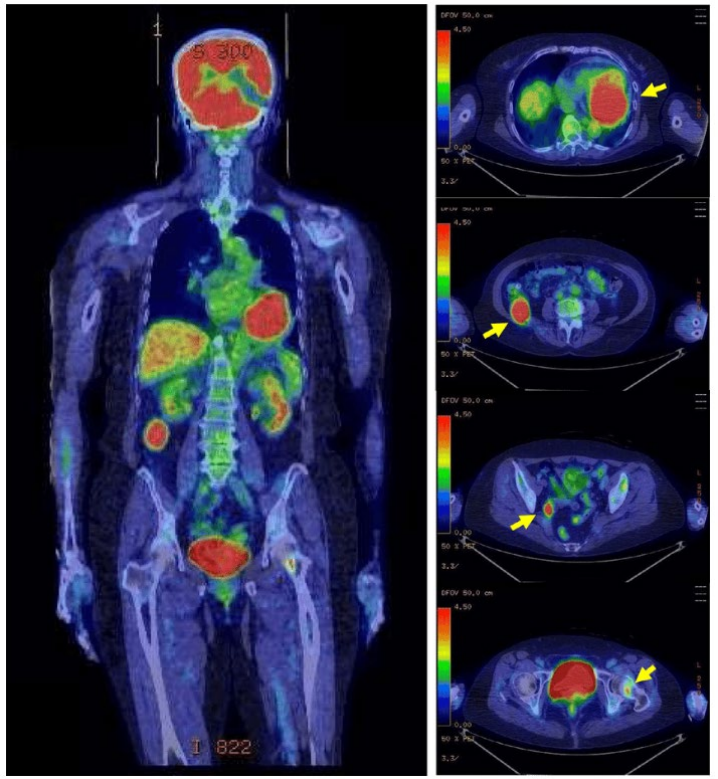
-
-



Example from the clinic
Positron Emission Tomography



The basic principle of a positron emission tomography (PET) system: A PET detector ring detects a pair of gamma photons with an energy of 511 keV (red arrows) which results from the annihilation of an electron with a positron emitted by the radiotracer (FDG).



Beta Decay – Exercise

Why can ^{60}Co be a β^-emitter?

Why can ^{57}Co not be a β^+emitter?

Beta Decay – Exercise

^{60}Co as β^- emitter: $\quad {}^{60}_{27}Co \rightarrow {}^{60}_{28}Ni + \beta^- + \bar{\nu}$

$$Q = (\hat{m}_{{}^{60}_{27}Co}) - (\hat{m}_{{}^{60}_{28}Ni} + m_e)$$
$$= \left(m_{{}^{60}_{27}Co} - 27\, m_e\right) - \left(m_{{}^{60}_{28}Ni} - 28\, m_e + m_e\right)$$
$$= m_{{}^{60}_{27}Co} - m_{{}^{60}_{28}Ni} = 59.933822 - 59.930791 > 0$$

Nuclear masses

Beta Decay – Exercise

^{57}Co as β^+emitter: $\quad {}^{57}_{27}Co \rightarrow {}^{57}_{26}Fe + \beta^+ + \nu$

$$Q = (\hat{m}_{{}^{57}_{27}Co}) - (\hat{m}_{{}^{57}_{26}Fe} + m_e)$$

$$= \left(m_{{}^{57}_{27}Co} - 27\, m_e\right) - \left(m_{{}^{57}_{26}Fe} - 26\, m_e + m_e\right)$$

Nuclear masses

$$= m_{{}^{57}_{27}Co} - m_{{}^{57}_{26}Fe} - 2m_e = 56.936\ldots - 56.935\ldots - 2 \cdot 0.000549 < 0$$

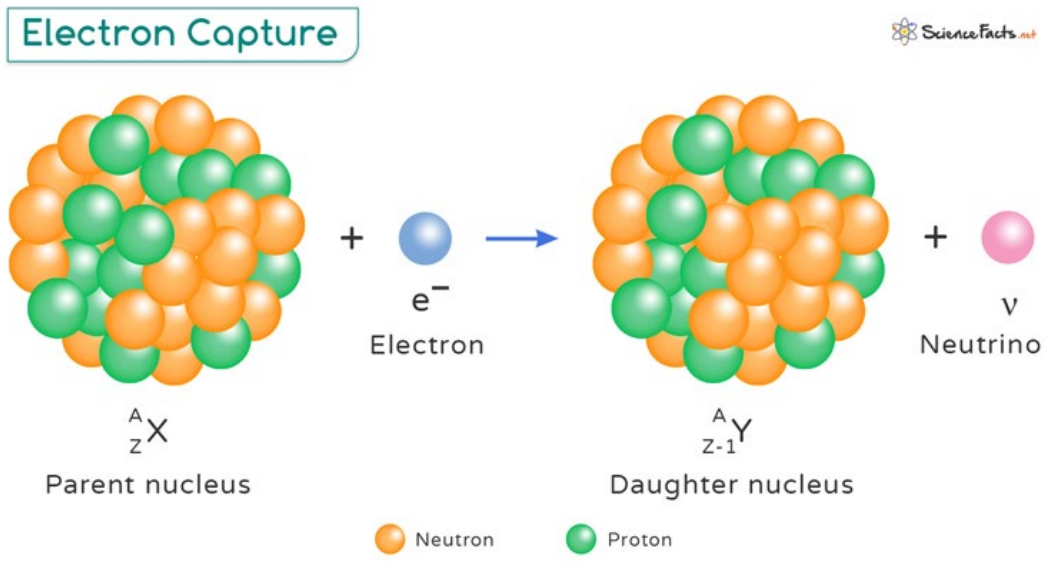
Beta Decay- electron capture (EC)

- A nuclear process that competes with positron emission is *electron capture (EC or $\mathcal{E}$)*

$$p + e^- \rightarrow n + \nu$$

$$ {}^{A}_{Z}X_N + e^- \rightarrow {}^{A}_{Z-1}X'_{N+1} + \nu $$

$$ Q = \left[m\left({}^{A}X\right) - m\left({}^{A}X'\right)\right]c^2 $$



TU/e

Electron capture

- Nucleus absorbs low-orbit (K-shell or L-shell) electron
- Vacancy is filled with free electron or electron from outer orbits
- Binding energy is used to emit high-energy photon, or eject other outer electrons (Auger-electrons)
- Nucleus (if in excited state after ejecting neutrino) returns to ground state by emitting gamma(s) or by IC (internal conversion: energy transmitted to electron in inner shell which is then ejected)

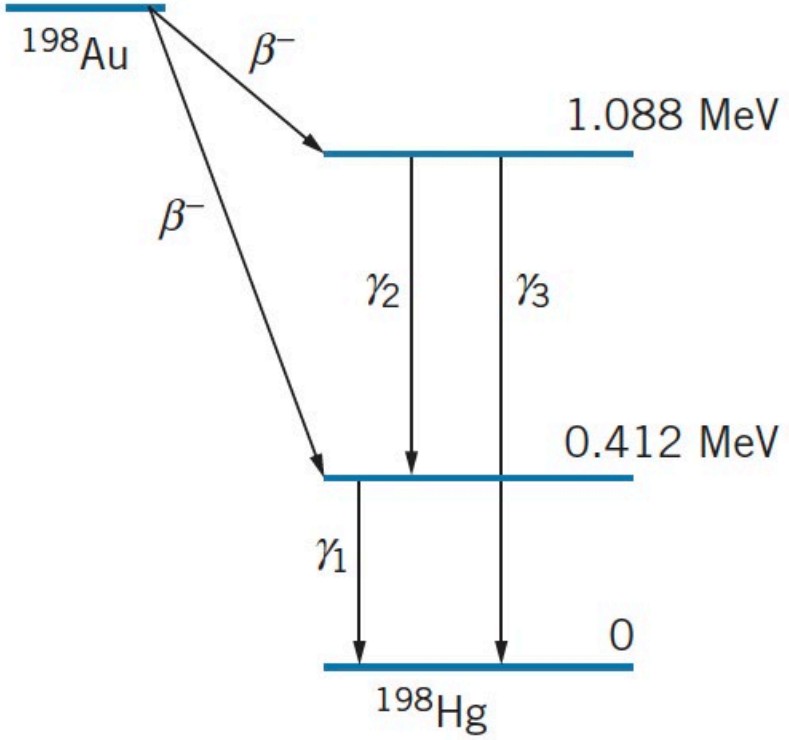
Gamma Decay

- Following alpha or beta decay, the final nucleus may be left in an excited state

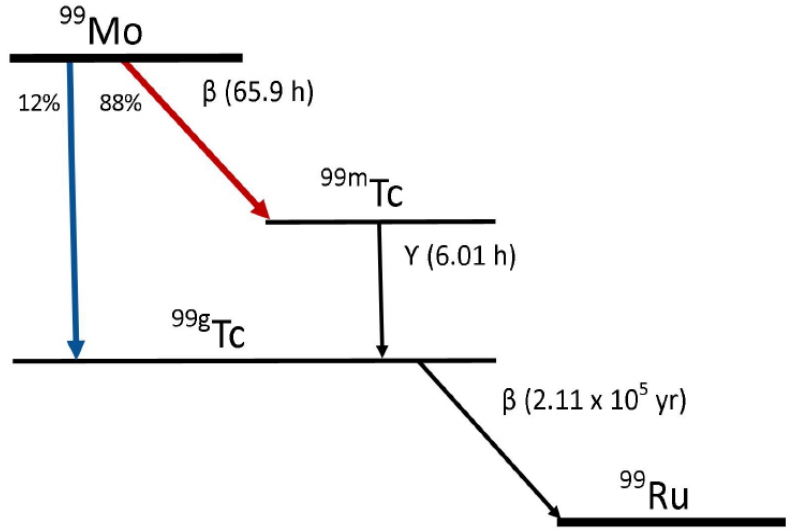
- Just as an atom does, the nucleus will reach its ground state after emitting one or more high-energy photons: gamma radiation

- Excited states with long lifetimes are called *metastable* (m), e.g. $^{99m}\mathrm{Tc}$ ($T_{1/2} = 6\ h$) and $^{81m}\mathrm{Kr}$ ($T_{1/2} = 13\ s$)

- Typical energies: 100 keV to few MeV
- Best stopped by high-density materials (lots of atoms per volume) with high Z (many electrons for interactions with gammas (photo-electric effect, Compton scattering)



- 99mTc (Technetium-99m) used primarily for diagnostic nuclear medicine
- widely used for SPECT (Single-Photon Emission Computed Tomography) images of the heart, brain, lungs, kidneys, bones, and other organs.
- Half-time of 6 h



- Rb/Kr: Rb ($T_{\frac{1}{2}Rb} = 4.5\ h, T_{\frac{1}{2}Kr83m} = 12s$)

TU/e

Gamma Decay – Examples from the Clinic

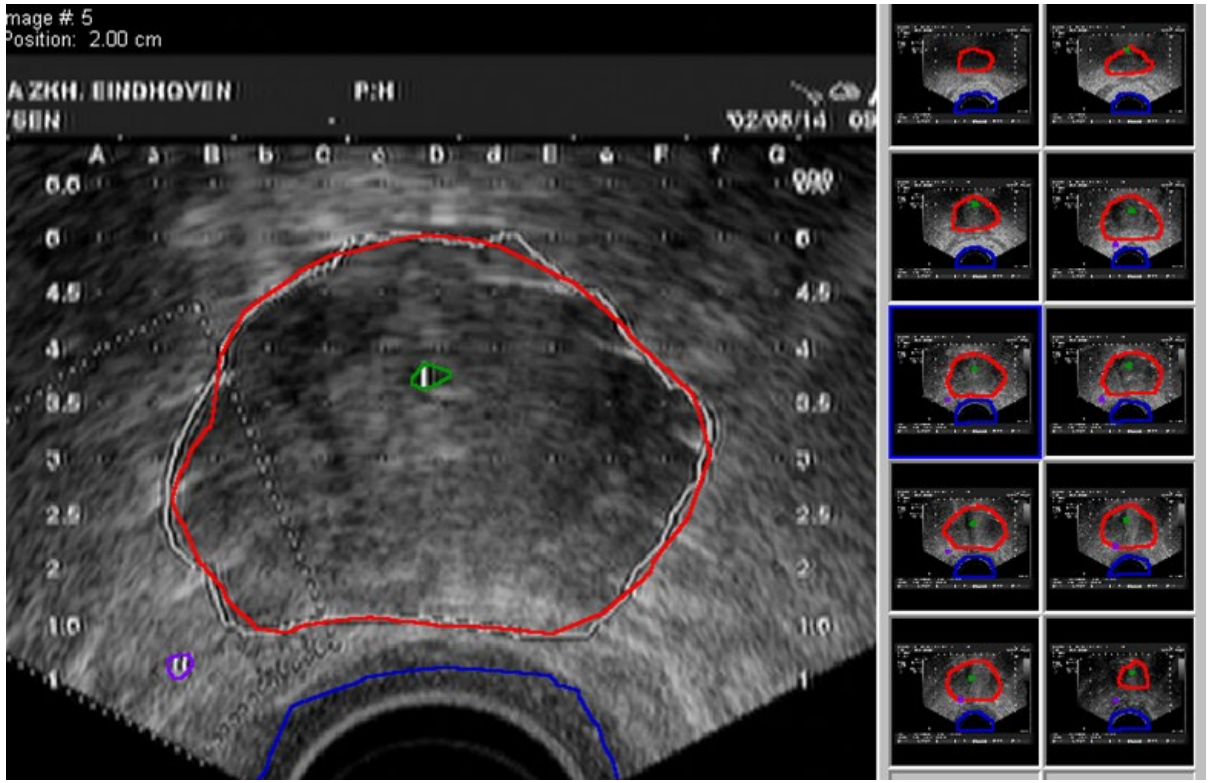


^{125}I implantation in the prostate

- Rectal probe guidance
- Hollow needles inserted by urologist

Slides and images from Catharina Hospital

Gamma Decay – Examples from the Clinic

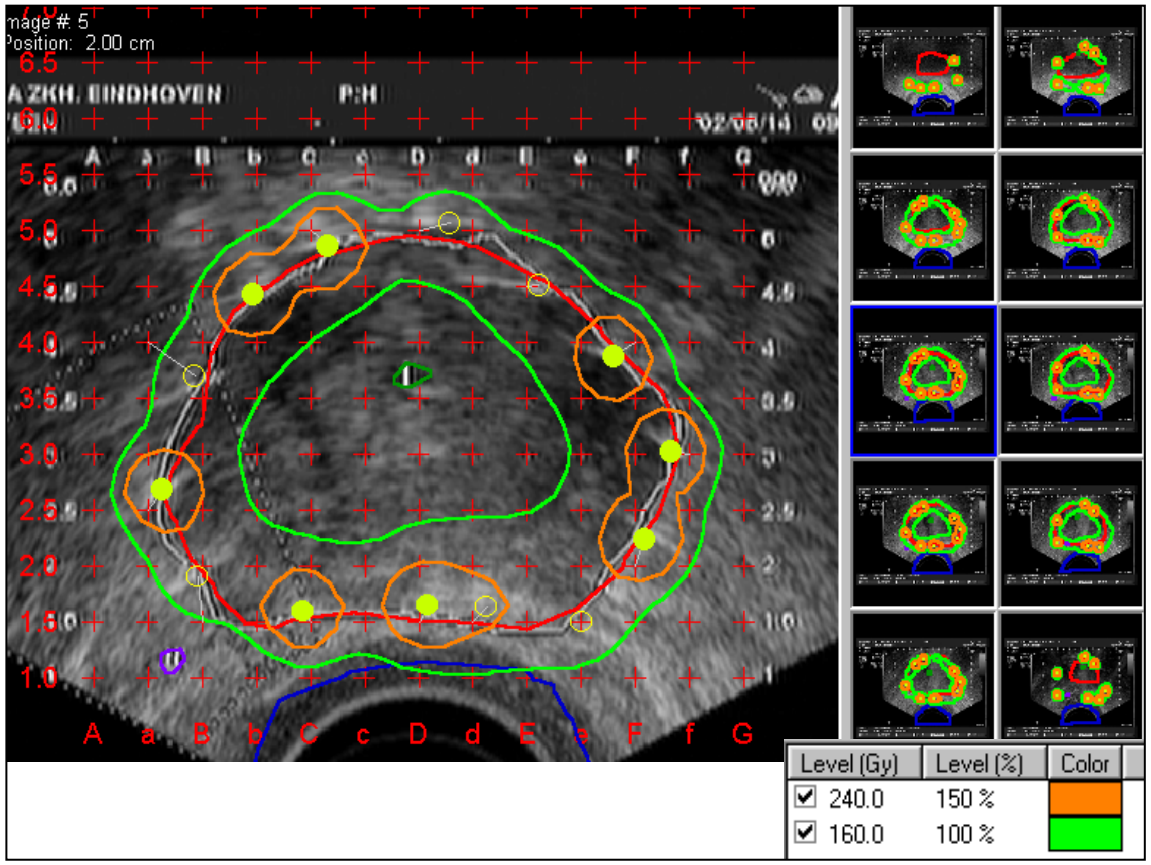


^{125}I implantation in the prostate

- Shape and position of needles are checked

Slides and images from Catharina Hospital

Gamma Decay – Examples from the Clinic



^{125}I implantation in the prostate

- Medical Physicist created irradiation plan online

Slides and images from Catharina Hospital

Conclusion

- Radioactive decay: unstable nucleus loses energy by radiation
- 3 types:
  - **Alpha:** A and Z change (He nucleus)
    stopped by sheet of paper
  - **Beta:** only Z changes (electron / positron / EC)
    stopped by some material (best: low-Z)
  - **Gamma:** no change in A and Z (photons)
    only stopped by a lot of metal (best: Pb)

Note:
Has to be positive in order for the decay to occur

$$Q = (m_{inital} - m_{final})c^2$$